WORKED SOLUTION

# STEP III 2017, Question 1

Pure • Telescoping Series Bounds

**STEP III 2017, Question 1**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q1

## The key idea

The identity in (i) writes each reciprocal binomial coefficient as a constant times a *difference* of consecutive terms, so the sum telescopes. In (ii), $1/n^3$ is awkward to sum but reciprocals of binomial coefficients are not: $\binom{n+1}{3}$ and $\binom{n+2}{5}$ are polynomials in $n$ that behave like multiples of $n^3$ and $n^5$, so we trap $1/n^3$ between expressions we can sum exactly using (i).

## Solution

### Part (i)

Write $\binom{m}{k} = \dfrac{m!}{k!\,(m-k)!}$. Then

$$\begin{aligned}\frac{1}{\binom{n+r-1}{r}} - \frac{1}{\binom{n+r}{r}} &= \frac{r!\,(n-1)!}{(n+r-1)!} - \frac{r!\,n!}{(n+r)!} \\ &= \frac{r!\,(n-1)!}{(n+r)!}[(n+r)-n] = \frac{r\cdot r!\,(n-1)!}{(n+r)!},\end{aligned}$$

where we took out the common factor $r!\,(n-1)!/(n+r)!$, using $(n+r)! = (n+r)\,(n+r-1)!$ and $n! = n\,(n-1)!$. Multiplying by $\frac{r+1}{r}$ gives

$$\frac{r+1}{r}\left(\frac{1}{\binom{n+r-1}{r}} - \frac{1}{\binom{n+r}{r}}\right) = \frac{(r+1)!\,(n-1)!}{(n+r)!} = \frac{1}{\binom{n+r}{r+1}},$$

since $(n+r)-(r+1) = n-1$. This is the required identity.

**The sum.** Put $a_n = 1/\binom{n+r-1}{r}$, so that the identity reads $1/\binom{n+r}{r+1} = \frac{r+1}{r}(a_n - a_{n+1})$. Summing from $n=1$ to $N$, everything cancels except the first and last terms:

$$\sum_{n=1}^{N} \frac{1}{\binom{n+r}{r+1}} = \frac{r+1}{r}(a_1 - a_{N+1}) = \frac{r+1}{r}\left(1 - \frac{1}{\binom{N+r}{r}}\right),$$

because $a_1 = 1/\binom{r}{r} = 1$. As $N \to \infty$, $\binom{N+r}{r} \geq N + r \to \infty$ (for $r \geq 1$), so

$$\sum_{n=1}^{\infty} \frac{1}{\binom{n+r}{r+1}} = \frac{r+1}{r}.$$

**The deduction.** With $r = 2$ the sum from $n = 1$ is $\frac{3}{2}$. The $n = 1$ term is $1/\binom{3}{3} = 1$, so

$$\sum_{n=2}^{\infty} \frac{1}{\binom{n+2}{3}} = \frac{3}{2} - 1 = \frac{1}{2}.$$

### Part (ii)

**First inequality.** For $n \geq 3$,

$$\frac{1}{\binom{n+1}{3}} = \frac{3!}{(n+1)n(n-1)} = \frac{3!}{n^3 - n} > \frac{3!}{n^3},$$

since $0 < n^3 - n < n^3$ (the left inequality because $n \geq 2$).

**Second inequality.** We have $\binom{n+1}{3} = \frac{1}{6}n(n^2-1)$ and $\binom{n+2}{5} = \frac{1}{120}(n+2)(n+1)n(n-1)(n-2) = \frac{1}{120}n(n^2-1)(n^2-4)$. So

$$\frac{20}{\binom{n+1}{3}} - \frac{1}{\binom{n+2}{5}} = \frac{120}{n(n^2-1)} - \frac{120}{n(n^2-1)(n^2-4)} = \frac{120\,(n^2-5)}{n(n^2-1)(n^2-4)}.$$

For $n \geq 3$ all the factors $n$, $n^2-1$, $n^2-4$ are positive, so this is less than $120/n^3 = 5!/n^3$ if and only if

$$n^2(n^2-5) < (n^2-1)(n^2-4), \quad \text{that is,} \quad n^4 - 5n^2 < n^4 - 5n^2 + 4,$$

which is true.

? **Ask yourself** Which sums from part (i) do we need, and where do they start?

We need $\sum_{n\geq 3} 1/\binom{n+1}{3}$ and $\sum_{n\geq 3} 1/\binom{n+2}{5}$. In the first, put $m = n-1$: it becomes $\sum_{m\geq 2} 1/\binom{m+2}{3} = \frac{1}{2}$, which is exactly the deduction in (i). In the second, put $m = n-2$: it becomes $\sum_{m\geq 1} 1/\binom{m+4}{5}$, which is (i) with $r = 4$, giving $\frac{5}{4}$.

All the series involved converge (they are positive and bounded by convergent ones, as the inequalities themselves show). Summing a strict inequality between convergent series term by term gives a strict inequality, because the sum of the positive differences is at least its first term, which is positive. Write $T = \sum_{n\geq 3} 1/n^3$.

**Upper bound.** Summing the first inequality over $n \geq 3$:

$$6T < \sum_{n=3}^{\infty} \frac{1}{\binom{n+1}{3}} = \frac{1}{2}, \qquad \text{so} \qquad T < \frac{1}{12}.$$

**Lower bound.** Summing the second inequality over $n \geq 3$:

$$120\,T > 20 \cdot \frac{1}{2} - \frac{5}{4} = \frac{35}{4}, \qquad \text{so} \qquad T > \frac{35}{480} = \frac{7}{96}.$$

Finally, $\sum_{n\geq 1} 1/n^3 = 1 + \frac{1}{8} + T$ and $1 + \frac{1}{8} = \frac{108}{96}$. Hence

$$\frac{108+7}{96} = \frac{115}{96} < \sum_{n=1}^{\infty} \frac{1}{n^3} < \frac{108}{96} + \frac{8}{96} = \frac{116}{96},$$

as required.

## Checks

- Test the identity with $n = r = 1$: $1/\binom{2}{2} = 1$ and $2\,(1/\binom{1}{1} - 1/\binom{2}{1}) = 2 \cdot \frac{1}{2} = 1$.
- With $r = 1$ the sum is $\sum 1/\binom{n+1}{2} = \sum \frac{2}{n(n+1)} = 2$, the familiar telescoping sum, matching $\frac{r+1}{r} = 2$.
- Numerically $\sum 1/n^3 \approx 1.20206$, while $\frac{115}{96} \approx 1.19792$ and $\frac{116}{96} \approx 1.20833$, so the bounds hold and pin the value down to within about 0.01.

## Common slips

- Getting the starting index wrong after shifting: $\sum_{n\geq 3} 1/\binom{n+1}{3}$ starts at $\binom{4}{3}$, which is the $n = 2$ term of the deduction in (i), not the $n = 1$ term.
- Forgetting the terms $1 + \frac{1}{8}$ for $n = 1, 2$, which the inequalities do not cover.

## Beyond STEP

The number $\sum 1/n^3$ is $\zeta(3)$, Apéry's constant; unlike $\sum 1/n^2 = \pi^2/6$ it has no known closed form, and Apéry's 1978 proof that it is irrational used rapidly convergent series not unlike the ones here. Comparing a series with a telescoping one, and summing inequalities term by term, are standard tools of a first analysis course. For a careful treatment of convergence and comparison tests, see Abbott, *Understanding Analysis*.

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